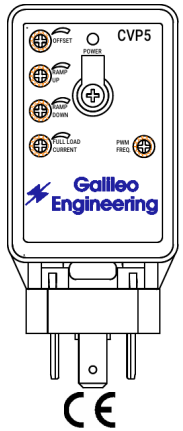
Galileo Engineering

CVP5 Electronic Driver for proportional valves

CVP5-H



General Description

This miniature electronic regulator is embedded into the plug housing and allows the open loop driving of proportional valve solenoid. It is protected against power supply polarity inversion and solenoid short circuit.
The minimum and maximum current values are adjusted with two potentiometers, and two other separate potentiometers allow the adjustment of ramp-up and ramp-down parameter. A yellow led is lit when the system is powered.

CVP5-H is assembled with a DIN43650 - ISO 4400 connector, CVP5-P comes with a M12 connector and is rated IP67 (preliminary).

Applications

1 - On-Off application mode with switch and ramp setting for acceleration and deceleration uses.

The **GND (4)** and **3** terminals are connected to the two terminals of the switch (normally open). When the switch is closed, the input reference signal is tied to the maximum voltage value and therefore the current of the solenoid reaches the maximum value. When the switch is open the current flowing into the solenoid reaches the minim value. The **ramp up** and **ramp down** potentiometers allow adjustment, using a linear ramp, of respectively the time delay between the transition from minimum to maximum current and the delay between the switching from maximum to minimum current. The minimum and maximum current values are adjusted with the **offset** and **full load** potentiometers.

2 - Control mode using a voltage generator as input signal

The external signal control must be connected to terminal **3** and ground (0V) must be connected to terminal **2**. The input voltage on the terminal **3** can be regulated from 0 to 10V. The current on the valve coil is proportional to the input command voltage. Set this signal to the maximum value (10V), then proceed to the adjustment of the **full load** potentiometer, in order to set the maximum current value on the solenoid.

3 - Control mode with potentiometer

Pins 1, 2 and 3 of the potentiometer must be connected respectively to the **GND (4), 3** and **2** terminals of the controller**.** To setup the controller, rotate the potentiometer fully clockwise and follow the **"Adjustment instructions".**
A 5KOhm potentiometer is recommended. In any case the potentiometer value must be between 2KOhm and 5KOhm.

4 - Two axes control with joystick

This control can be done using a dual axis joystick and two CVP5 devices.

Adjustment instructions

After the system is connected, verify the correct movement of the hydraulic cylinder using the potentiometer or the switch.

Set the **ramp up** and **ramp down** potentiometers to zero, rotating the cursor completely counterclockwise.

Set the external potentiometer to zero (or set the the external switch to the 'open' position) and set the minimum current of the solenoid using the **offset** potentiometer, by rotating it until the hydraulic device begins to move. This setting allows the system to operate without delay.

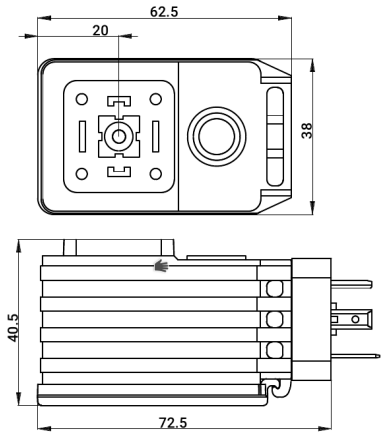
Set the **full load** potentiometer to zero and rotate the external control potentiometer completely clockwise (or set the external switch to the 'closed' position). Then, next rotate the full load potentiometer clockwise until the hydraulic cylinder reaches the maximum displacement, then rotate the full load potentiometer back until the hydraulic cylinder comes back slightly.

Once the tuning of the start and end positions of the hydraulic cylinder stroke is complete, it is possible to regulate the switching speed between the two extreme positions of the stroke, using the **ramp up** and **ramp down** potentiometers. This further adjustment will not affect the previously tuned settings.

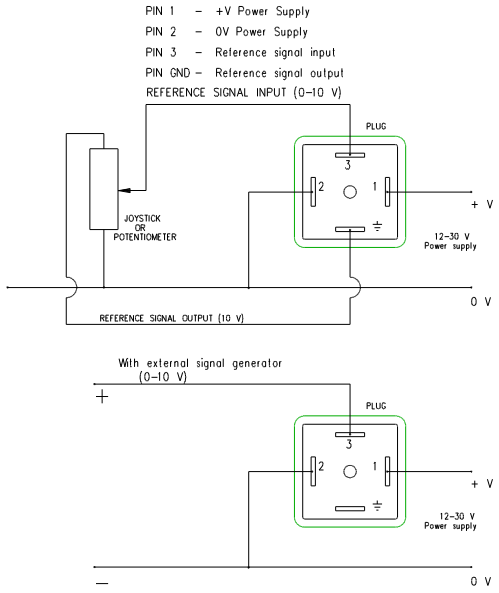
Technical Specifications

Power Supply Voltage (abs. Min-max)	12-30Vdc
Rectified and filtered ripple voltage (max)	10%
Output Current (adjustable) (min-max)	0 – 2.0A
Current absorption (max) without load	30mA
Offset current (adjustable) (min-max)	0 – 0.6A
Input voltage reference range	0-10V
Maximum absorbed power (average)	35W
PWM Frequency (adjustable)	100 – 500Hz
Adjustable Ramp down time	0.1 – 10s
Adjustable Ramp up time	0.1 – 10s
Current stability (on temp. range)	3%
Operating temperature range	-20 ... +70 °C
Protection class CVP5-H	IP65
Weight	100g

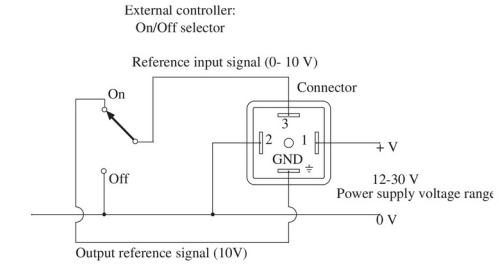
Dimensions



Electrical Connections



Electrical connections (continued)



Note

The GND pin is otherwise marked as pin 4 inside the controller.

The power supply voltage must be in the 12 to 30 V dc range. It is necessary to power the system with rectified and filtered voltage. The use of a 4700 F 35V electrolytic capacitor is recommended in order to filter the power voltage supply.

The electronic controller can drive valves with coil powered at 12 or 24 Vdc. To assure the nominal maximum current value of the coil the voltage supply of the controller must exceed the nominal voltage supply of the coil valve at least of 1,5V.